UNIVERSITY OF CENTRAL FLORIDA
Center for the Study of Human Trafficking and Modern Slavery
9907 Universal Boulevard
Orlando, FL 32819-8701

November 8, 2025

RE: - Letter of Recommendation – Harper Archambault

To Whom It May Concern,

It is my great pleasure to recommend Harper Archambault for her exceptional contributions to the development of the AI Firewall component of our "Buyer Beware" project, conducted in partnership with the UCF Center for the Study of Human Trafficking and Modern Slavery (HTMS Center). Harper completed this work as part of her Computer Science Senior Design course (COP 4935) from January 2025 through November 2025.

Throughout the project, Harper demonstrated outstanding technical skills, innovative thinking, and a deep sense of social responsibility. She played an integral role in designing and programming the AI Firewall, a system capable of detecting, filtering, and flagging online activity patterns associated with human trafficking recruitment, solicitation, and exploitation. Her approach combined advanced machine learning techniques, natural language processing, and behavioral analytics to create a system that aligns with real-world law enforcement and industry applications.

Harper's ability to translate complex ethical challenges into effective technological solutions distinguished her from her peers. She approached the project not merely as a coding assignment but as a humanitarian mission balancing technical precision with a commitment to protecting vulnerable populations. She collaborated seamlessly across multidisciplinary teams, integrating feedback hospitality researchers and survivor-informed advisors. Her attention to data privacy, algorithmic fairness, and responsible AI practices ensured that the project's outcomes met both academic rigor and ethical integrity.

Specific Contributions:

- Harper worked on the networking side of the Buyer Beware project, where she developed the real-time web scraper that collects data for the AI analysis pipeline and can be deployed on the Raspberry Pi.
- Over the course of the project, she tested several scraping methods to find an approach that could extract as much relevant text as possible while maintaining fast performance and full compatibility with the Raspberry Pi architecture and ultimately designed a Python-based scraping system that captures and processes web content in real time using asynchronous requests, custom parsers, and text-cleaning modules.
- The scraper leverages Redis for efficient queue management, enabling continuous data streaming to the AI system for analysis. It was built to balance speed with thoroughness, using concurrent processing and adaptive parsing to handle a wide range of web sources effectively.
- She also conducted extensive testing to ensure the scraper's accuracy, reliability, and scalability under different network conditions.

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The Buyer Beware project represents a milestone in the HTMS Center's pursuit of AI-driven prevention strategies, and Harper's leadership and ingenuity have been pivotal to its success. I am confident that her combination of technical excellence, social awareness, and collaborative spirit will make her an outstanding contributor to any advanced research or professional setting she chooses to pursue.

Please feel free to contact me at JWB@ucf.edu should you require further information regarding her contributions or the project's outcomes.

Regards in Hospitality,

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